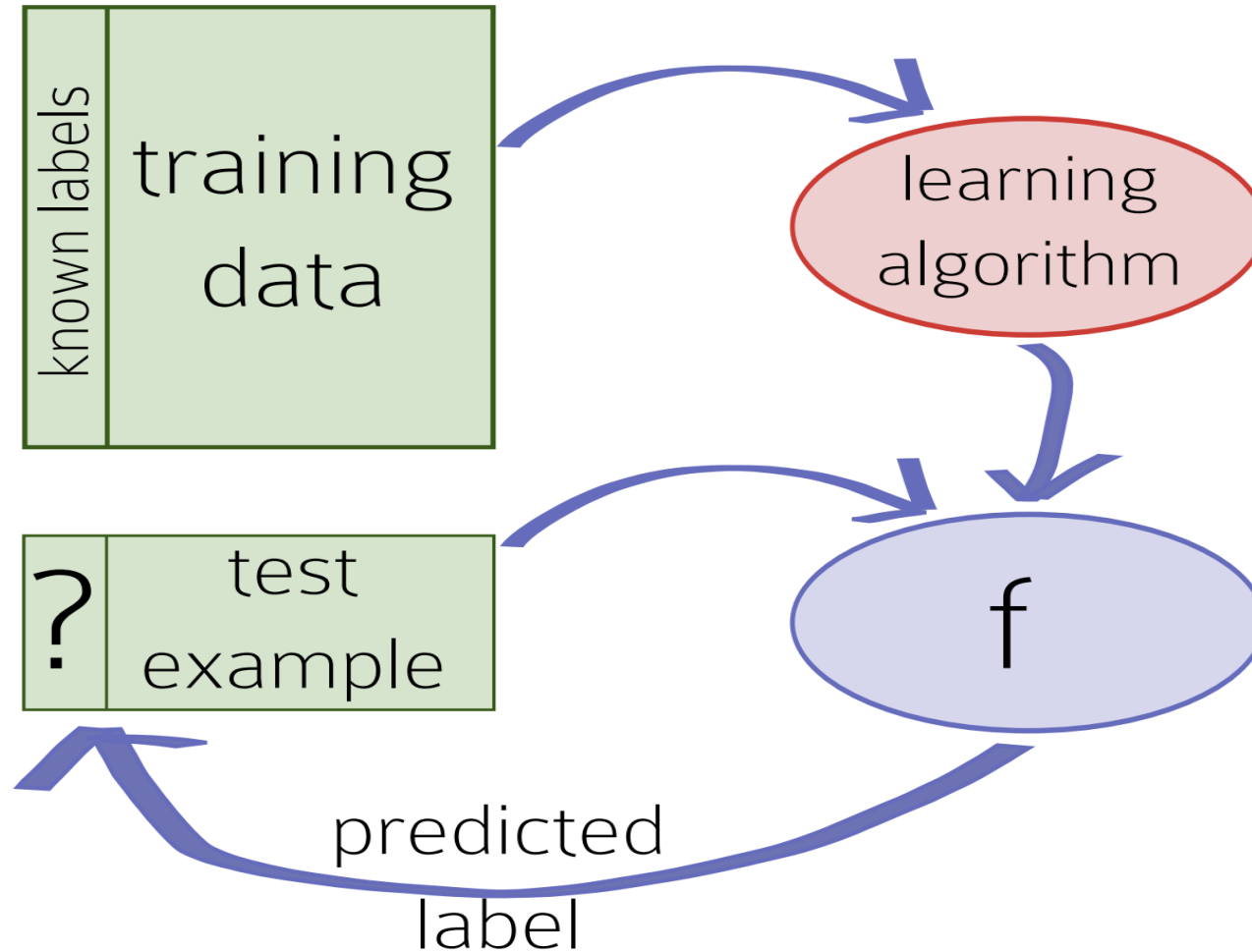


Supervised Learning

- Supervised learning is the types of machine learning in which machines are trained using well "labelled" training data, and on basis of that data, machines predict the output. The labelled data means some input data is already tagged with the correct output.
- In supervised learning, the training data provided to the machines work as the supervisor that teaches the machines to predict the output correctly. It applies the same concept as a student learns in the supervision of the teacher.
- Supervised learning is a process of providing input data as well as correct output data to the machine learning model. The aim of a supervised learning algorithm is to **find a mapping function to map the input variable(x) with the output variable(y).**

Supervised approach to Machine Learning



Classification: Decision Trees

Introduction

- Classic and natural model of learning
- Based on fundamental computer science notion of “divide and conquer”
- It creates a model that predicts the value of a target variable based on several input variables
- A tree can be learned by splitting the source set into subsets based on an attribute value test using recursive partitioning
- The recursion is completed when the subset at a node has all the same value of the target variable, or when splitting no longer adds value to the predictions.
- Although decision trees can be applied to many learning problems, we will begin with the simplest case: binary classification

The decision tree model of learning

Do people like movies of Amitabachan?

The decision tree model of learning

Do people like movies of Amitabachan?

- ▶ Q : Is the genre of movie is “Action”?
 - ▶ Yes
- ▶ Q : Is the lead actor is Amitabachan?
 - ▶ Yes
- ▶ Q : Has people liked previous Action movie of the Amitabachan?
 - ▶ Yes

People do like Action movies of Amitabachan

The decision tree model of learning

Do people like movies of Amitabachan?

- ▶ Q : Is the genre of movie is “Comedy”?
 - ▶ Yes
- ▶ Q : Is the lead actor is Amitabachan?
 - ▶ Yes
- ▶ Q : Has people liked previous comedy movie of the Amitabachan?
 - ▶ No

People don't like Comedy movies of Amitabachan

The decision tree model of learning

Goal of learner: Figure out what questions to ask, and in what order, and what to predict when you have answered enough questions

The decision tree model of learning

Lead Actor	Genre	Hit(Y/N)
Amitabh Bacchan	Action	Yes
Amitabh Bacchan	Fiction	Yes
Amitabh Bacchan	Romance	No
Amitabh Bacchan	Action	Yes
<i>Abhishek Bacchan</i>	<i>Action</i>	<i>No</i>
<i>Abhishek Bacchan</i>	<i>Fiction</i>	<i>No</i>
<i>Abhishek Bacchan</i>	<i>Romance</i>	<i>Yes</i>

The decision tree model of learning

Lead Actor	Genre	Hit(Y/N)
Amitabh Bacchan	Action	Yes
Amitabh Bacchan	Fiction	Yes
Amitabh Bacchan	Romance	No
Amitabh Bacchan	Action	Yes
<i>Abhishek Bacchan</i>	<i>Action</i>	<i>No</i>
<i>Abhishek Bacchan</i>	<i>Fiction</i>	<i>No</i>
<i>Abhishek Bacchan</i>	<i>Romance</i>	<i>Yes</i>

- Questions are features
- Responses are feature values
- Decision is the **label**

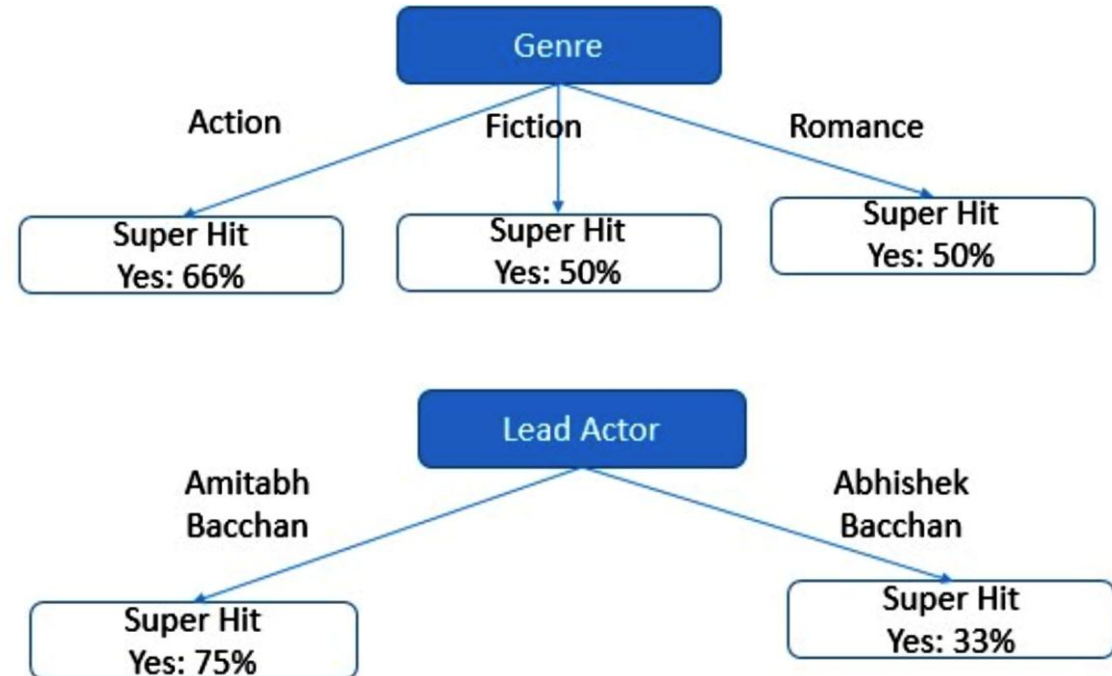
Why it's Decision Tree?

Lead Actor	Genre	Hit(Y/N)
Amitabh Bacchan	Action	Yes
Amitabh Bacchan	Fiction	Yes
Amitabh Bacchan	Romance	No
Amitabh Bacchan	Action	Yes
<i>Abhishek Bacchan</i>	<i>Action</i>	<i>No</i>
<i>Abhishek Bacchan</i>	<i>Fiction</i>	<i>No</i>
<i>Abhishek Bacchan</i>	<i>Romance</i>	Yes

Why it's Decision Tree?

Lead Actor	Genre	Hit(Y/N)
Amitabh Bacchan	Action	Yes
Amitabh Bacchan	Fiction	Yes
Amitabh Bacchan	Romance	No
Amitabh Bacchan	Action	Yes
Abhishek Bacchan	Action	No
Abhishek Bacchan	Fiction	No
Abhishek Bacchan	Romance	Yes

- We can represent set of questions and guesses in a tree format



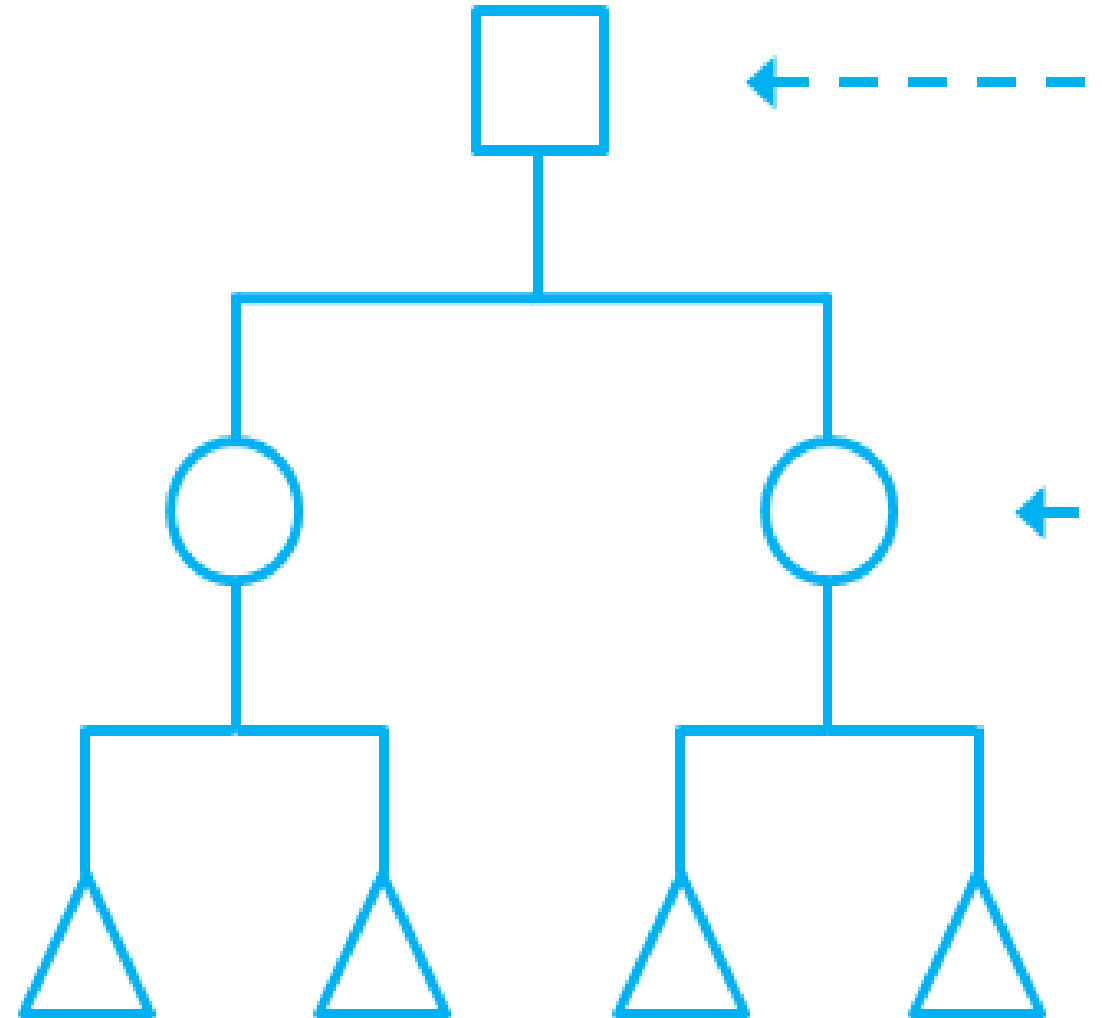
Terminologies

Root Node: is the beginning of a tree

Internal Node: splits into further nodes

Leaf Node: is a node that no longer splits

Branch: is the link between nodes



Classification—A Two-Step Process

- **Model construction**: describing a set of predetermined classes
 - Each tuple/sample is assumed to belong to a predefined class, as determined by the **class label attribute**
 - The set of tuples used for model construction is **training set**
 - The model is represented as classification rules, decision trees, or mathematical formulae
- **Model usage**: for classifying future or unknown objects
 - **Estimate accuracy** of the model
 - The known label of test sample is compared with the classified result from the model
 - **Accuracy** rate is the percentage of test set samples that are correctly classified by the model
 - **Test set** is independent of training set (otherwise overfitting)
 - If the accuracy is acceptable, use the model to **classify new data**
- Note: If *the test set* is used to select models, it is called **validation (test) set**

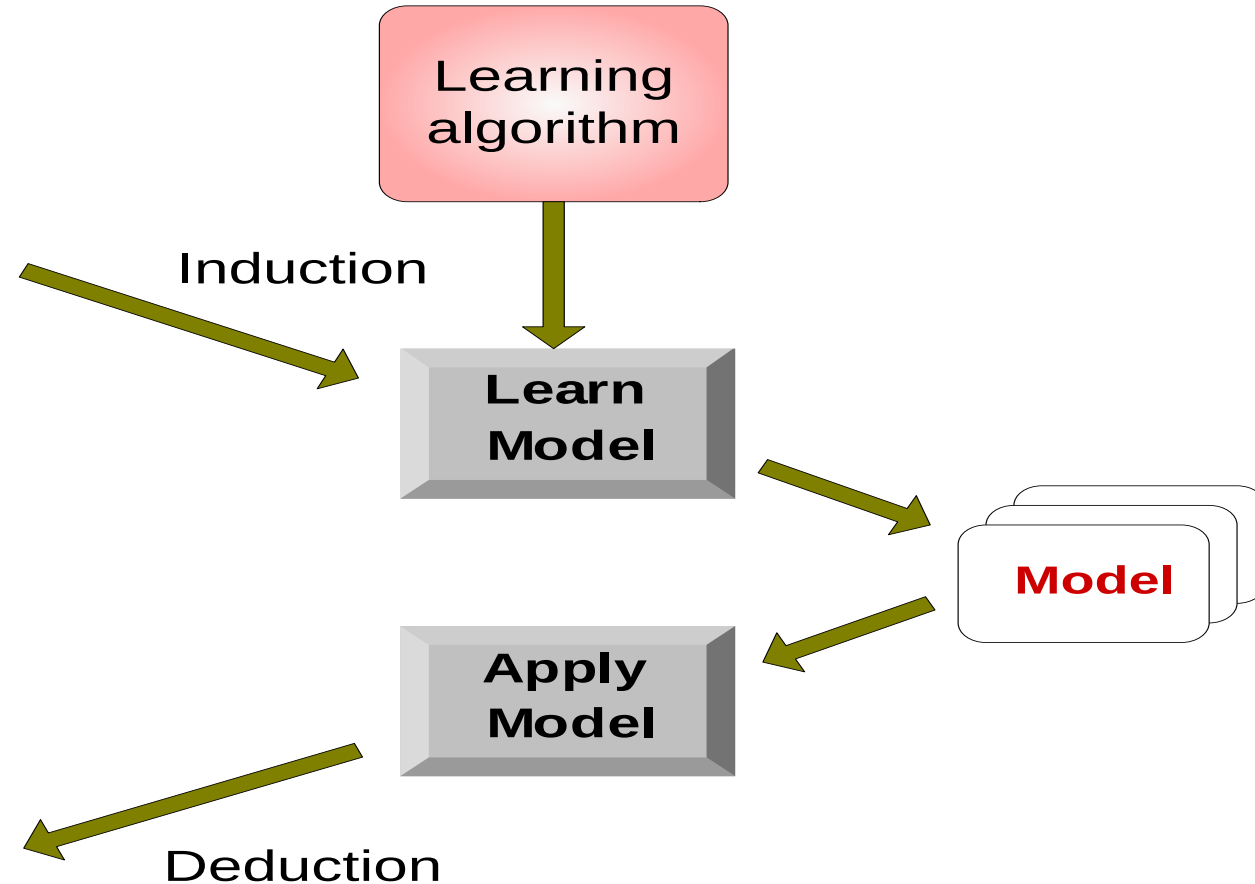
Illustrating Classification Task

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Training Set

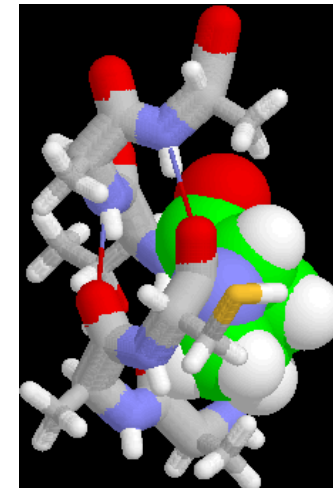
<i>Tid</i>	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

Test Set



Examples of Classification Task

- Predicting tumor cells as benign or malignant
- Classifying credit card transactions as legitimate or fraudulent
- Classifying secondary structures of protein as alpha-helix, beta-sheet, or random coil
- Categorizing news stories as finance, weather, entertainment, sports, etc



Example of a Decision Tree

<i>Tid</i>	<i>Refund</i>	<i>Marital Status</i>	<i>Taxable Income</i>	<i>Cheat</i>
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

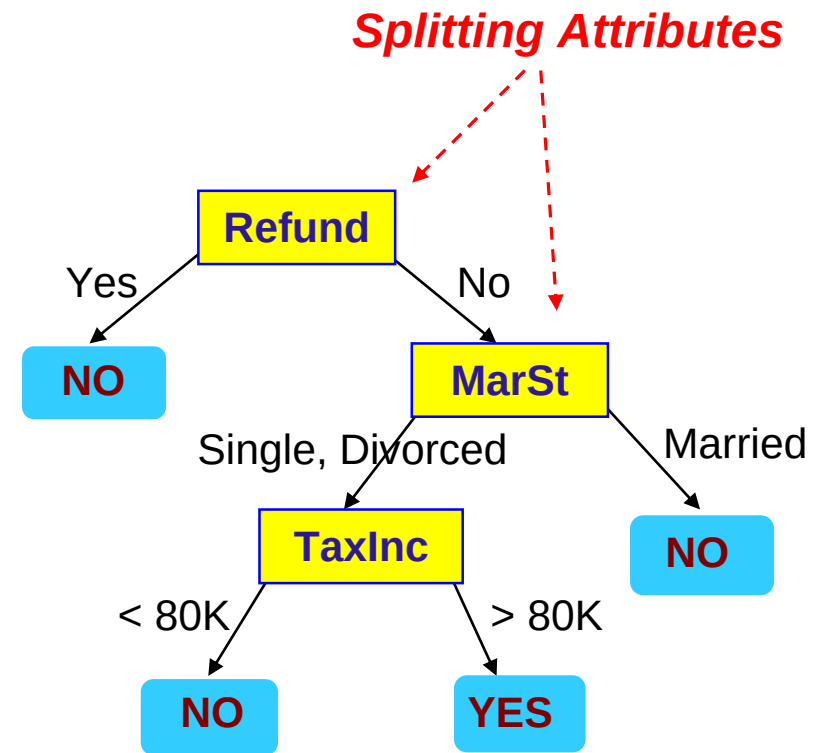
categorical

categorical

continuous

class

Training Data

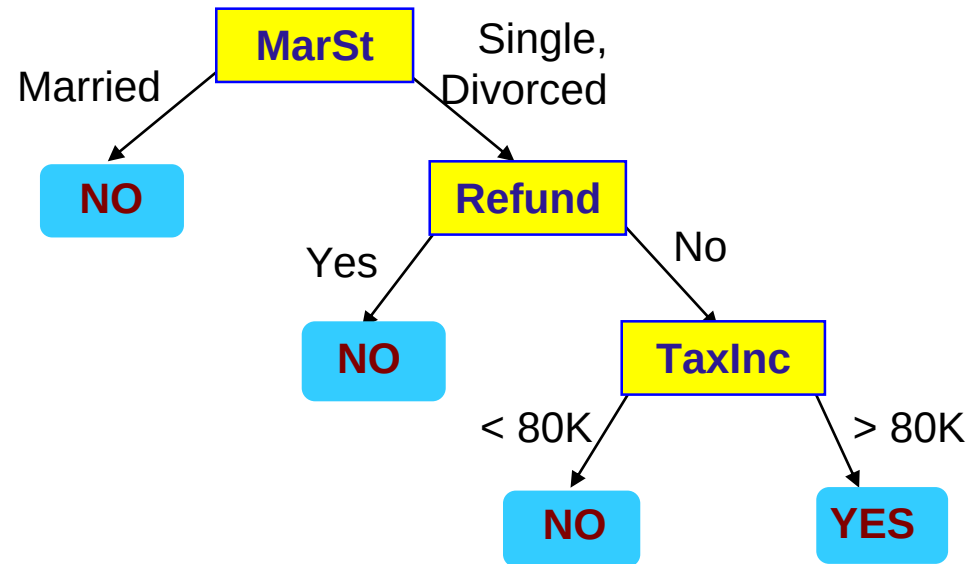


Model: Decision Tree

Another Example of Decision Tree

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

categorical
categorical
continuous
class



There could be more than one tree that fits the same data!

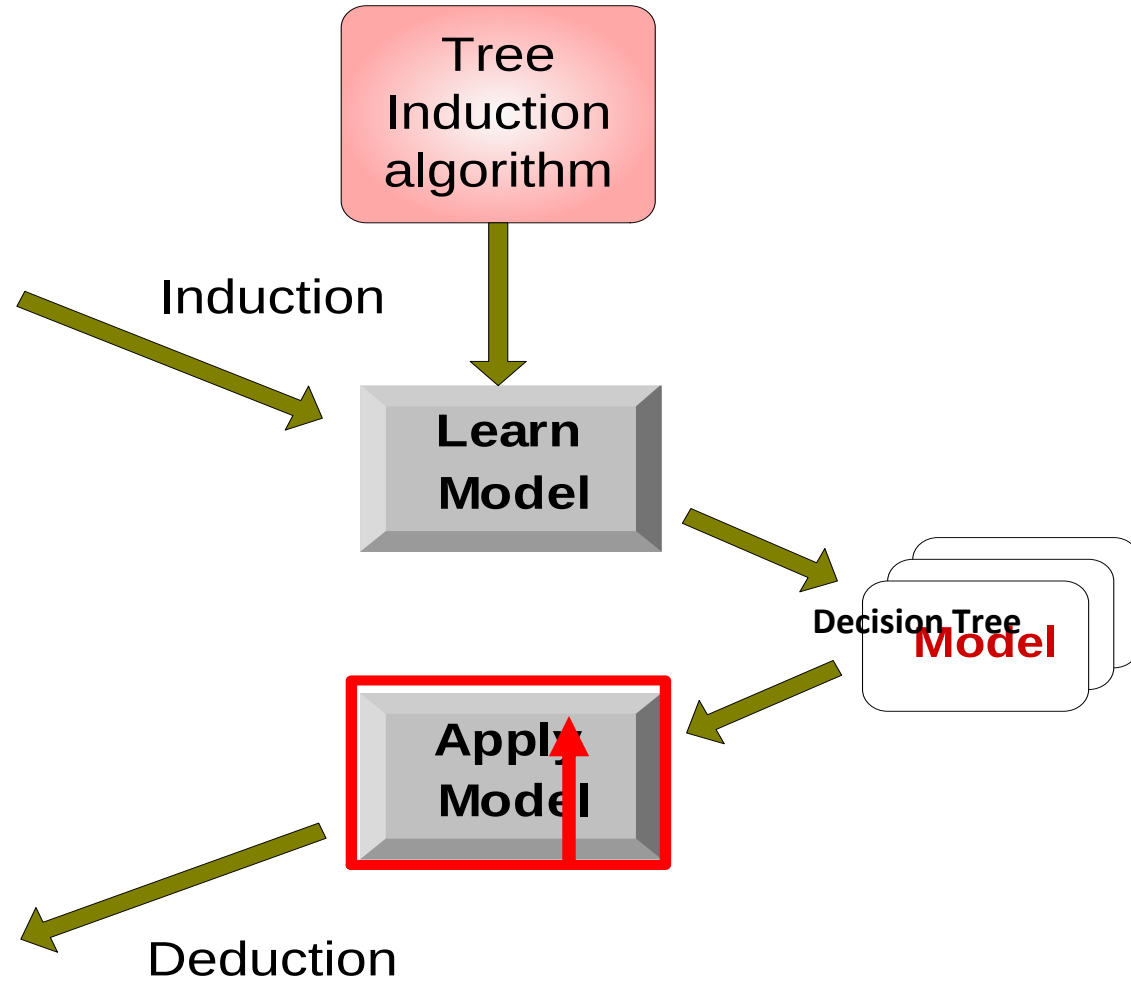
Decision Tree Classification Task

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Training Set

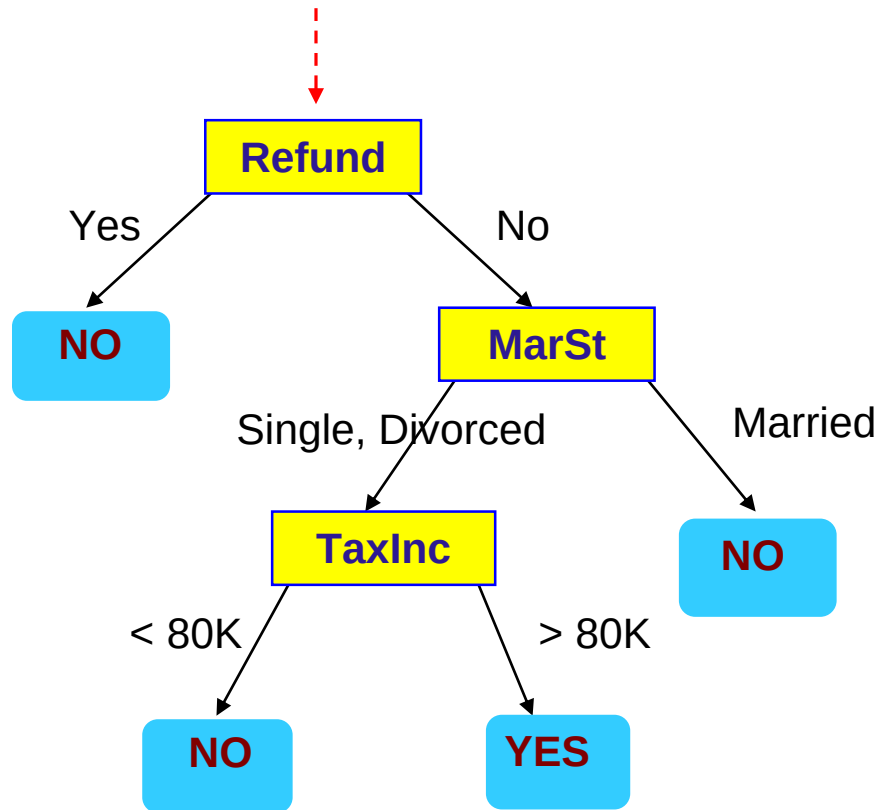
Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

Test Set



Apply Model to Test Data

Start from the root of tree.



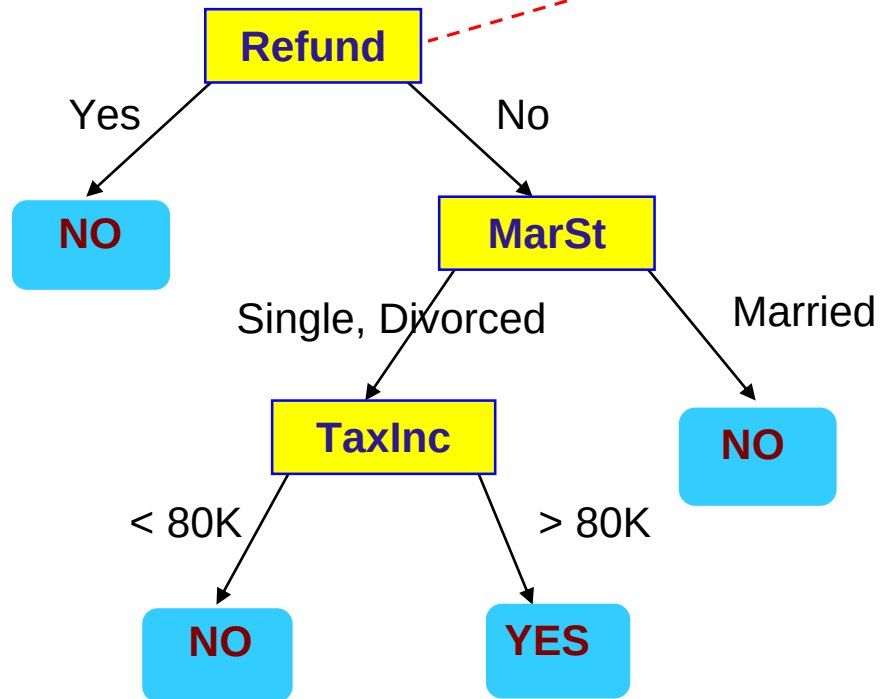
Test Data

Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?

Apply Model to Test Data

Test Data

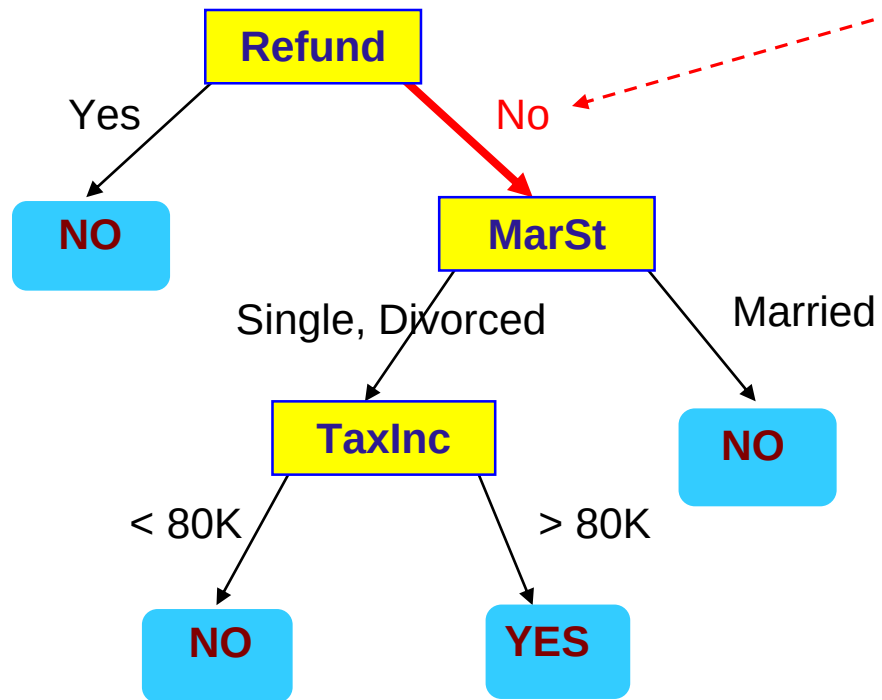
Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



Apply Model to Test Data

Test Data

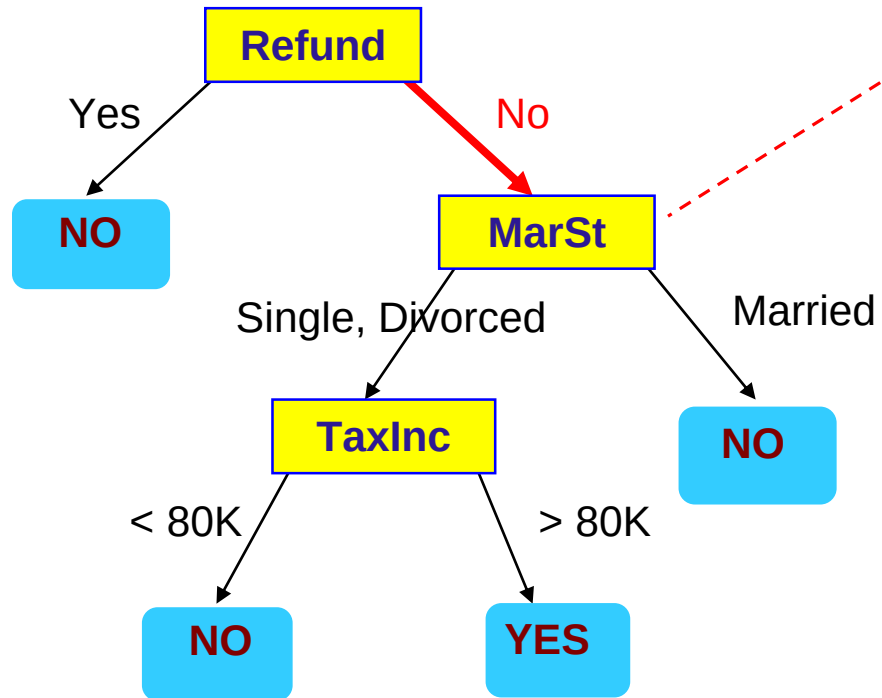
Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



Apply Model to Test Data

Test Data

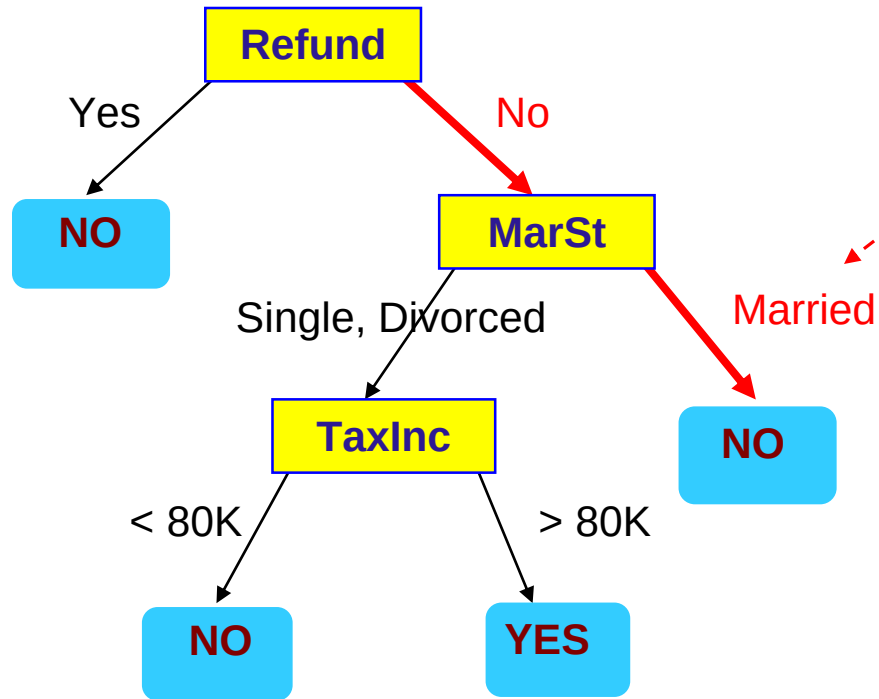
Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



Apply Model to Test Data

Test Data

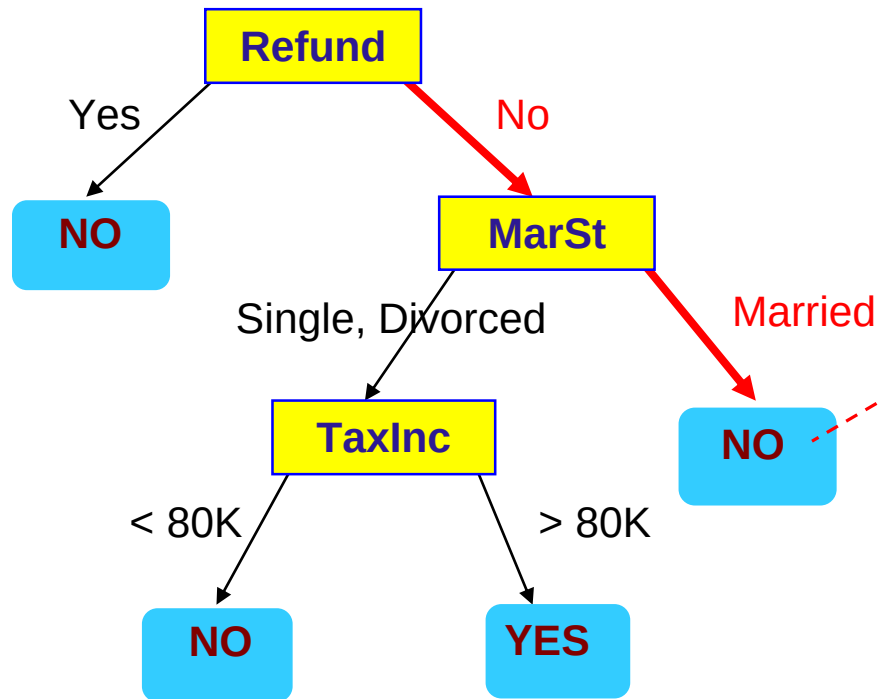
Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



Apply Model to Test Data

Test Data

Refund	Marital Status	Taxable Income	Cheat
No	Married	80K	?



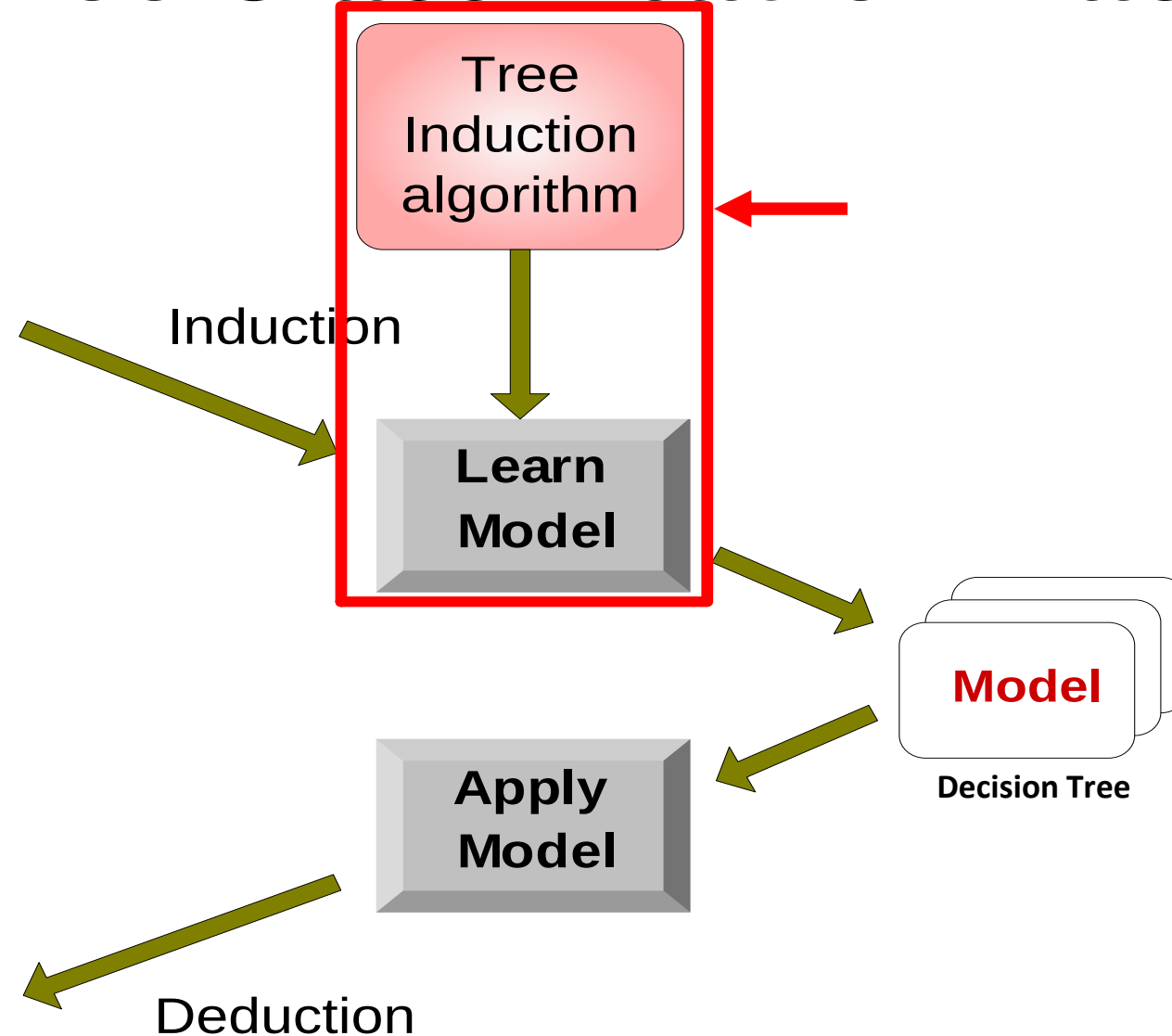
Assign Cheat to "No"

Decision Tree Classification Task

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Training Set

Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?



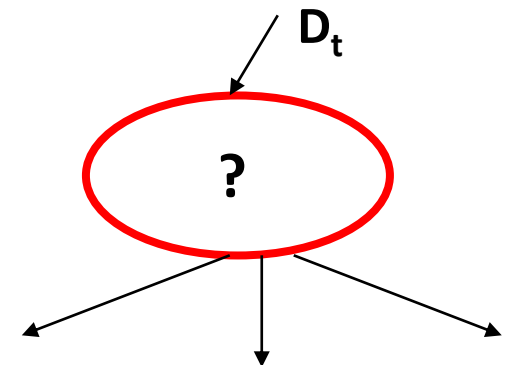
Some DT algorithms

- Hunt's Algorithm (one of the earliest)
- ID3 (Iterative Dichotomiser 3)
- C4.5, 5.0
- CART (Classification And Regression Trees)
- CHAID (CHi-squared Automatic Interaction Detector)
- MARS (Multivariate Adaptive Regression Splines)
- Conditional Inference Trees

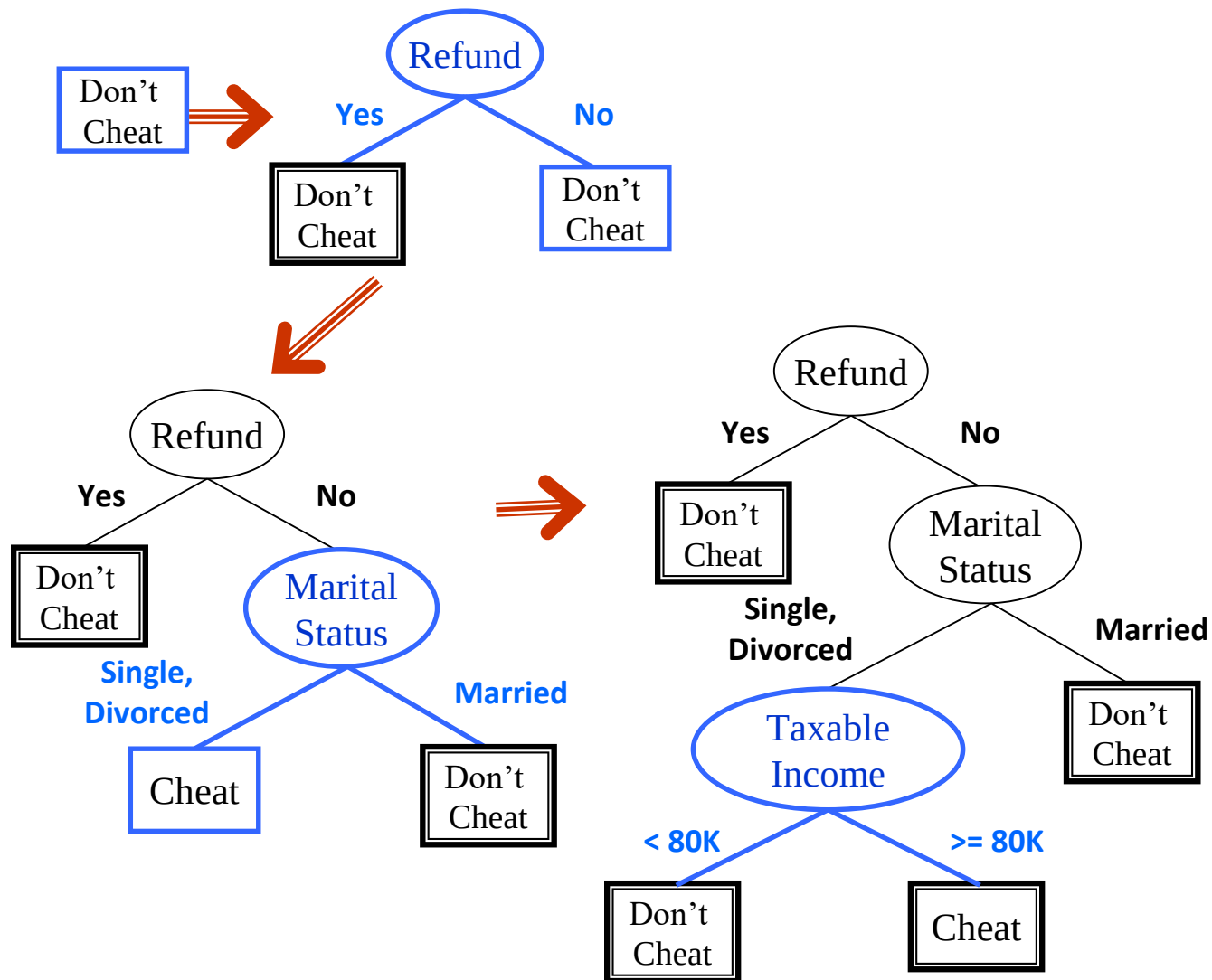
General Structure of Hunt's Algorithm

- Let D_t be the set of training records that reach a node t
- General Procedure:
 - If D_t contains records that belong the same class y_t , then t is a leaf node labeled as y_t
 - If D_t is an empty set, then t is a leaf node labeled by the default class, y_d
 - If D_t contains records that belong to more than one class, use an attribute test to split the data into smaller subsets. Recursively apply the procedure to each subset.
- This Splitting procedure is what can vary for different tree learning algorithms

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes



Hunt's Algorithm



Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Working Idea

At any node t:

- 1.If all records belong to same class → make **leaf node**
- 2.If no attributes left → assign **majority class**
- 3.Else:
 1. Select a feature
 2. Split data
 3. Recursively apply to subsets

- It **does NOT specify how to choose the best attribute**
It is just a **generic tree-building procedure**

Pseudocode Idea

```
Hunt(D):  
  if all samples same class:  
    return leaf  
  else:  
    choose attribute A  
    split D into subsets  
    for each subset:  
      apply Hunt recursively
```

Tree Induction

- Greedy strategy.
 - Split the records based on an attribute test that optimizes certain criterion.
- Issues
 - Determine how to split the records
 - How to specify the attribute test condition?
 - How to determine the best split?
 - Determine when to stop splitting

Tree Induction

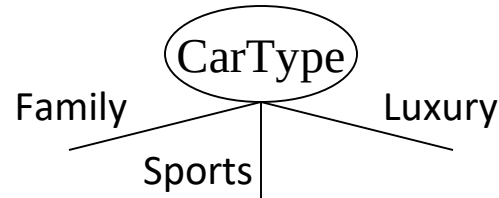
- Greedy strategy.
 - Split the records based on an attribute test that optimizes certain criterion.
- Issues
 - Determine how to split the records
 - How to specify the attribute test condition?
 - How to determine the best split?
 - Determine when to stop splitting

How to Specify Test Condition?

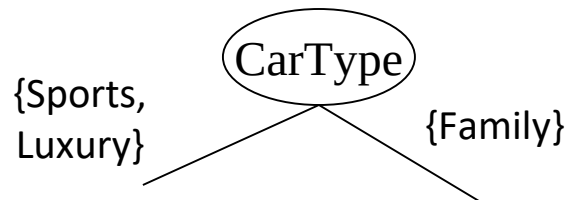
- Depends on attribute types
 - Nominal
 - Ordinal
 - Continuous
- Depends on number of ways to split
 - 2-way split
 - Multi-way split

Splitting Based on Nominal Attributes

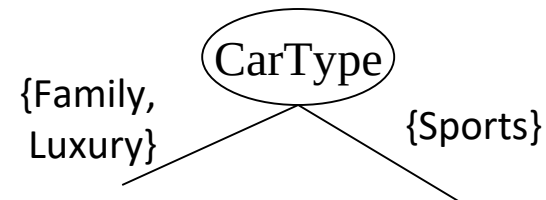
- **Multi-way split:** Use as many partitions as distinct values.



- **Binary split:** Divides values into two subsets.
Need to find optimal partitioning.

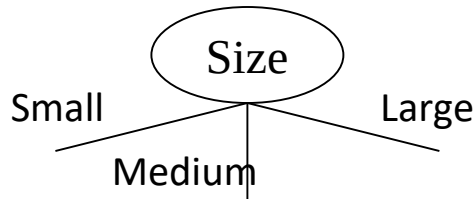


OR

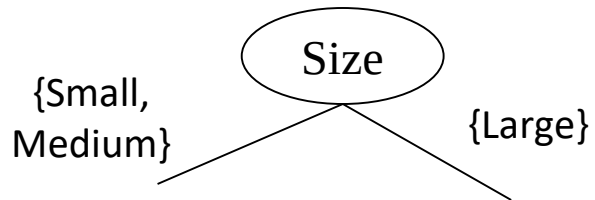


Splitting Based on Ordinal Attributes

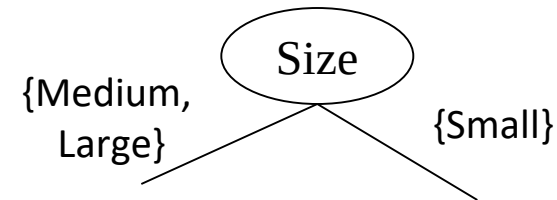
- **Multi-way split:** Use as many partitions as distinct values.



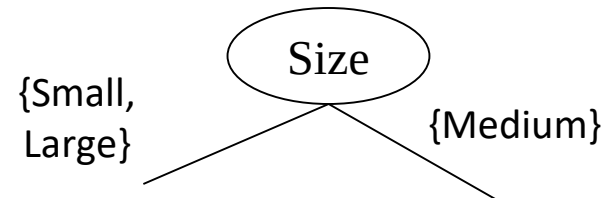
- **Binary split:** Divides values into two subsets.
Need to find optimal partitioning.



OR



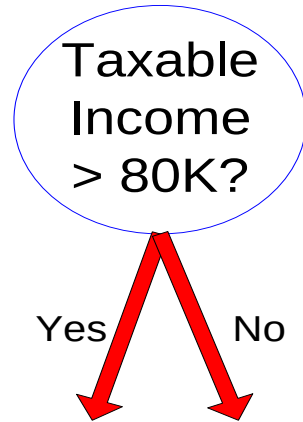
- What about this split?



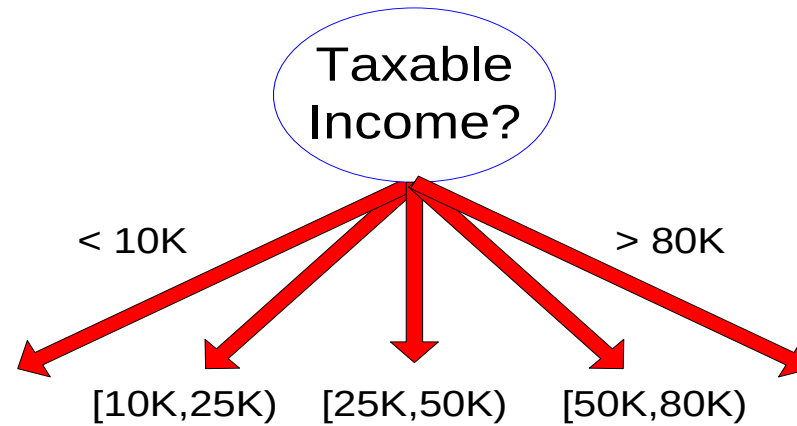
Splitting Based on Continuous Attributes

- Different ways of handling
 - **Discretization** to form an ordinal categorical attribute
 - Static – discretize once at the beginning
 - Dynamic – ranges can be found by equal interval bucketing, equal frequency bucketing (percentiles), or clustering.
 - **Binary Decision**: $(A < v)$ or $(A \geq v)$
 - consider all possible splits and finds the best cut
 - can be more compute intensive

Splitting Based on Continuous Attributes



(i) Binary split



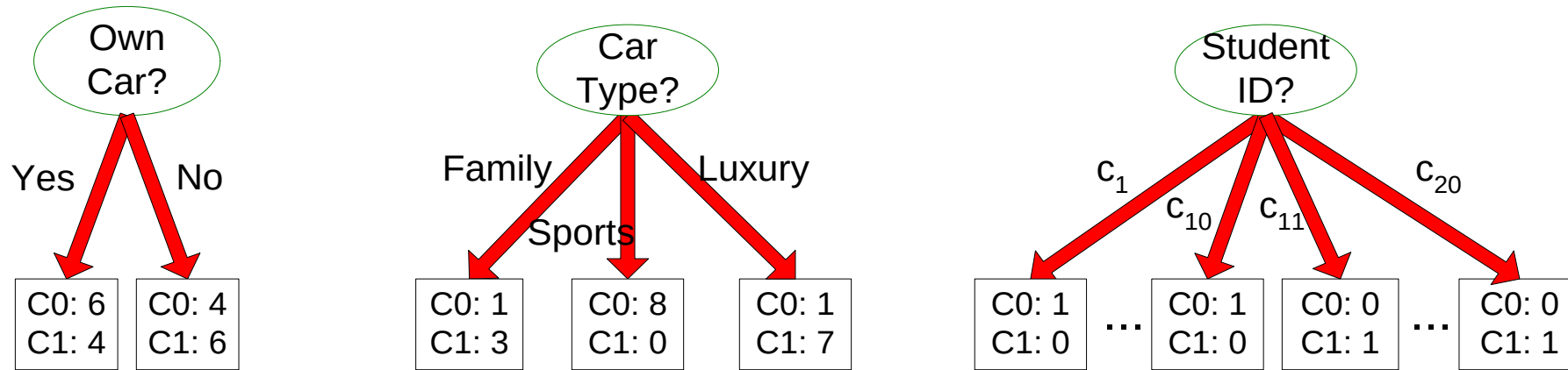
(ii) Multi-way split

Tree Induction

- Greedy strategy.
 - Split the records based on an attribute test that optimizes certain criterion.
- Issues
 - Determine how to split the records
 - How to specify the attribute test condition?
 - How to determine the best split?
 - Determine when to stop splitting

How to determine the Best Split

**Before Splitting: 10 records of class 0,
10 records of class 1**



Which test condition is the best?

How to determine the Best Split

- Greedy approach:
 - Nodes with **homogeneous** class distribution are preferred
- Need a measure of node impurity:

C0: 5
C1: 5

**Non-homogeneous,
High degree of impurity**

C0: 9
C1: 1

**Homogeneous,
Low degree of impurity**

Measures of Node Impurity

- Entropy
- Gini Index
- Misclassification error

Stopping Criteria for Tree Induction

- Stop expanding a node when all the records belong to the same class
- Stop expanding a node when all the records have similar attribute values

DT by Variable Type

- Single Variable Decision Trees

- Classifications are done with asking questions over only one variable

- Hybrid Decision Trees

- Classifications are done with asking questions over both single and multiple variables

- Multiple Variables Decision Trees

- Classifications are done with asking questions over multiple variables

ID3

- Developed by J. Ross Quinlan in 1979
- Quinlan was a computer science researcher in data mining, and decision theory.
- Received doctorate in computer science at the University of Washington in 1968.
- **ID3 (Iterative Dichotomiser 3)** is a **specific implementation** of decision tree learning.
- Based on Entropy
- Uses Information Gain to select the best feature
- Only works for discrete data
- Can not work with defective data
- Advantage over Hunt's algorithm is choosing the right attribute while classification. (Hunt's algorithm chooses randomly)



Steps

- Compute entropy
- Compute information gain for each attribute
- Choose attribute with maximum IG
- Split dataset
- Repeat recursively

Attribute Selection Measure: Information Gain (ID3) – Impurity Measure Entropy

1. Input: Dataset S with features and class labels
2. If all samples belong to same class
→ Return leaf node with that class
3. If no features left
→ Return majority class
4. Compute entropy of dataset
6. Select attribute with maximum IG
7. Split dataset based on that attribute
8. Recursively apply ID3 on each subset
9. Stop when nodes become pure

$$H(S) = - \sum p_i \log_2 p_i$$

5. For each attribute A :
 - Compute Information Gain

$$IG(S, A) = H(S) - \sum \frac{|S_v|}{|S|} H(S_v)$$

- Entropy = **impurity measure**
- Information Gain = **reduction in entropy**
- Best split → **maximum IG**

Entropy = Impurity Measure

◆ Entropy measures how impure or uncertain a dataset is

Situation	Entropy	Meaning
All same class	0	Pure (no uncertainty)
50–50 split	1 (max)	Maximum impurity
Mixed unequal	Between 0 and 1	Some uncertainty

Information Gain = Reduction in Entropy

Information Gain (IG) measures how much uncertainty is reduced after splitting data.

Before split:

Entropy = 0.971

After split:

Entropy = 0

- Before split → entropy = high
- After split → entropy = lower
- Difference = **Information Gain**

$$IG = 0.971 - 0 = 0.971$$

Big reduction → good split

Outlook	Humidity	Play
Sunny	High	No
Sunny	High	No
Overcast	High	Yes
Rain	High	Yes
Rain	Normal	Yes

Step 1: Compute Entropy of Dataset

Yes = 3

No = 2

$$H(S) = - \left(\frac{3}{5} \log_2 \frac{3}{5} + \frac{2}{5} \log_2 \frac{2}{5} \right) = 0.971$$

Step 2: Compute Information Gain for Each Attribute

Attribute 1: Outlook

Split subsets

Sunny → (No, No) → Entropy = 0

Overcast → (Yes) → Entropy = 0

Rain → (Yes, Yes) → Entropy = 0

$$P(\text{Yes}) = 0, \quad P(\text{No}) = 1$$

$$\begin{aligned} H(S) &= -(p_1 \log_2 p_1 + p_2 \log_2 p_2) \\ &= -(1 \cdot \log_2 1 + 0 \cdot \log_2 0) \end{aligned}$$

$$H(S) = -(1 \cdot 0 + 0) = 0$$

Weighted entropy

$$= \frac{2}{5}(0) + \frac{1}{5}(0) + \frac{2}{5}(0) = 0$$

Information Gain

$$IG(S, \text{Outlook}) = 0.971 - 0 = 0.971$$

Attribute 2: Humidity

Split subsets

High → (No, No, Yes, Yes)

- Yes = 2, No = 2
- Entropy = 1

Normal → (Yes)

- Entropy = 0

Weighted entropy

$$= \frac{4}{5}(1) + \frac{1}{5}(0) = 0.8$$

Information Gain

$$IG(S, \text{Humidity}) = 0.971 - 0.8 = 0.171$$

Step 3: Choose Best Attribute

Compare IG:

- Outlook = **0.971**
- Humidity = **0.171**

Select **Outlook**

Step 4: Create Root Node

Sunny $\rightarrow$ subset = (No, No) $\rightarrow$ pure $\rightarrow$ leaf = No

Overcast $\rightarrow$ (Yes) $\rightarrow$ leaf = Yes

Rain $\rightarrow$ (Yes, Yes) $\rightarrow$ leaf = Yes

All subsets are pure $\rightarrow$ stop recursion

```
      Outlook
     /  |  \
Sunny Overcast Rain
   No      Yes   Yes
```

Example 2

<i>Tid</i>	<i>Refund</i>	<i>Marital Status</i>	<i>Taxable Income</i>	<i>Class</i>
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No

Computing Information-Gain for Continuous-Valued Attributes

- Let attribute A be a continuous-valued attribute
- Must determine the *best split point* for A
 - Sort the value A in increasing order
 - Typically, the midpoint between each pair of adjacent values is considered as a possible *split point*
 - $(a_i + a_{i+1})/2$ is the midpoint between the values of a_i and a_{i+1}
 - The point with the *minimum expected information requirement* for A is selected as the split-point for A
- Split:
 - D1 is the set of tuples in D satisfying $A \leq \text{split-point}$, and D2 is the set of tuples in D satisfying $A > \text{split-point}$

Example 3

<i>RID</i>	<i>age</i>	<i>income</i>	<i>student</i>	<i>credit_rating</i>	<i>Class: buys_computer</i>
1	youth	high	no	fair	no
2	youth	high	no	excellent	no
3	middle_aged	high	no	fair	yes
4	senior	medium	no	fair	yes
5	senior	low	yes	fair	yes
6	senior	low	yes	excellent	no
7	middle_aged	low	yes	excellent	yes
8	youth	medium	no	fair	no
9	youth	low	yes	fair	yes
10	senior	medium	yes	fair	yes
11	youth	medium	yes	excellent	yes
12	middle_aged	medium	no	excellent	yes
13	middle_aged	high	yes	fair	yes
14	senior	medium	no	excellent	no

Example 4

Attribute				Classes
Gender	Car Ownership	Travel Cost	Income Level	Transportation
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Cheap	Medium	Train
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car

Example 5

Lead Actor	Genre	Hit(Y/N)
Amitabh Bacchan	Action	Yes
Amitabh Bacchan	Fiction	Yes
Amitabh Bacchan	Romance	No
Amitabh Bacchan	Action	Yes
<i>Abhishek Bacchan</i>	<i>Action</i>	<i>No</i>
<i>Abhishek Bacchan</i>	<i>Fiction</i>	<i>No</i>
<i>Abhishek Bacchan</i>	<i>Romance</i>	Yes

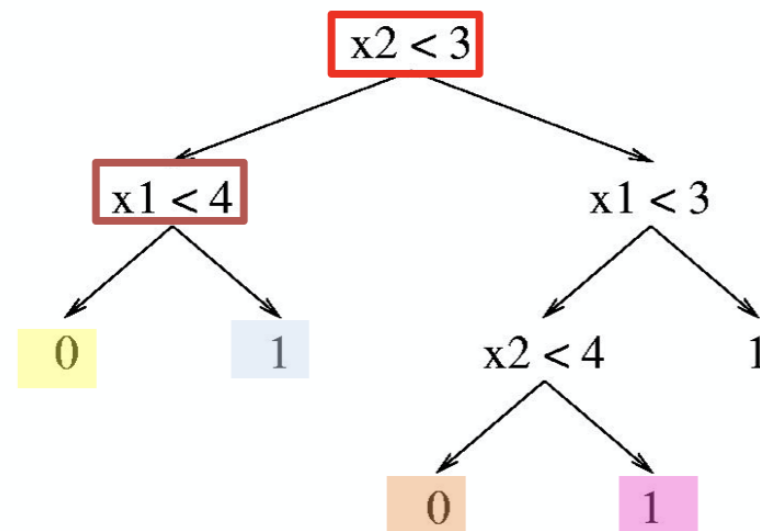
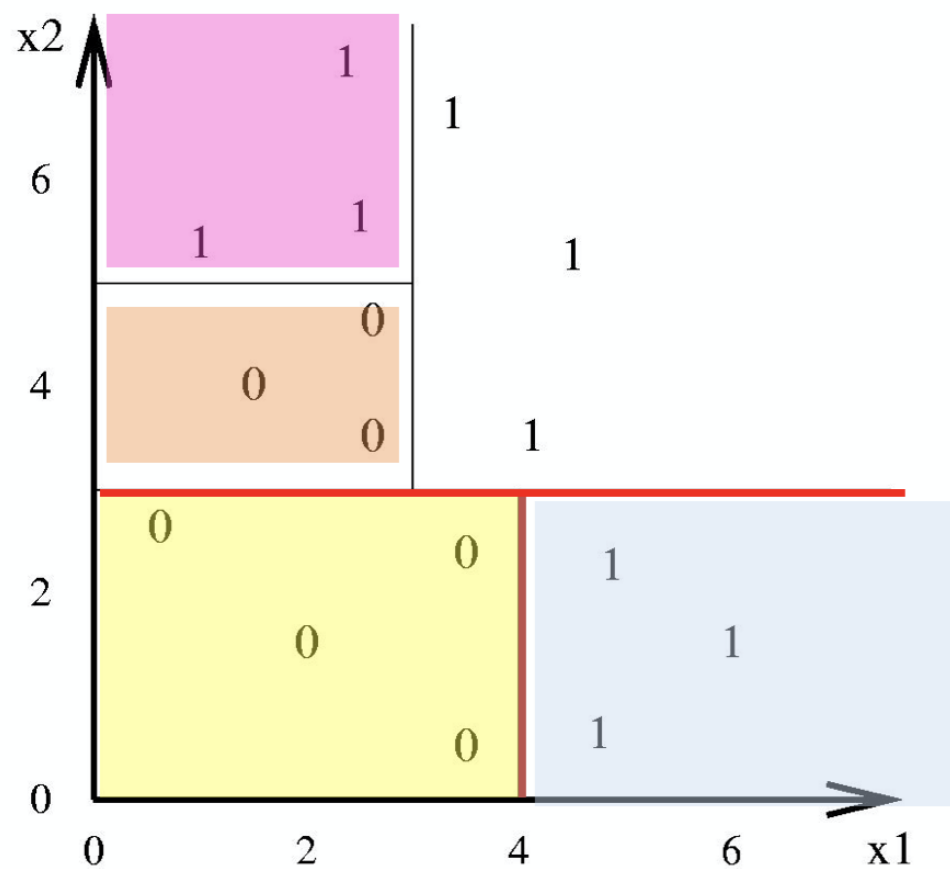
Pros of ID3 Algorithm

- Builds decision tree in minimum steps
- Each level benefits from previous level choices
- Whole dataset is scanned to create tree

Cons of ID3 Algorithm

- Tree can not be updated when new data is classified incorrectly, instead a new tree must be generated.
- Only one attribute at a time is tested for making a decision.
- Can not work with defective data
- Can not work with numerical attributes

Decision Tree Boundary



Decision Tree Boundary

